Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment No.: 6

Title: Sorting & Filtering Data

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Problem Statement

A sales manager wants to identify the company's highest-value transactions and products that are underperforming. Sorting and filtering in Power BI are used to examine transaction values and product performance and present useful business information.

Methodology / Steps Followed

Step 1: Data Collection

· Used the provided sales_analysis_dataset.xlsx workbook for the sales analysis assignment.

· The workbook contains Sales_Data for individual transactions and Product_Summary for product-level performance.

Step 2: Data Import

· Opened Power BI Desktop and selected Get Data → Excel workbook.

· Imported the Sales_Data and Product_Summary sheets for transaction-level and product-level analysis.

Step 3: Data Preparation

· Opened Transform Data to inspect the tables in Power Query Editor and reviewed the column names and data types.

· Checked Column Quality. The visible columns in the provided screenshot show 100% valid values, 0% errors, and 0% empty values.

· Reviewed the Date column and the Changed Type step before carrying out the sorting and filtering tasks.



	Transaction ID	Date	Region	Salesperson	Customer	Product	Category
1	TXN-1001	45932	East	Marcus Hill	Nova Pharma	Analytics Platform	Software
2	TXN-1002	45934	West	Daniel Cho	Meridian Auto	Smart Display 55in	Hardware
3	TXN-1003	45938	West	Daniel Cho	Nova Pharma	Analytics Platform	Software
4	TXN-1004	45938	South	Priya Nair	Juniper Foods	Wireless Router W2	Hardware
5	TXN-1005	45942	East	Grace Liu	Lumen Telecom	Business Laptop L5	Hardware
6	TXN-1006	45942	North	Tom Becker	Keystone Edu	Analytics Platform	Software
7	TXN-1007	45944	East	Grace Liu	Meridian Auto	Business Laptop L5	Hardware
8	TXN-1008	45945	East	Marcus Hill	Orion Travel	Business Laptop L5	Hardware
9	TXN-1009	45945	North	Tom Becker	Keystone Edu	Wireless Router W2	Hardware
10	TXN-1010	45946	South	Priya Nair	Apex Corp	Wireless Router W2	Hardware
11	TXN-1011	45947	West	Sofia Rossi	Delta Retail	Wireless Router W2	Hardware
12	TXN-1012	45950	East	Marcus Hill	Fusion Media	Analytics Platform	Software
13	TXN-1013	45952	East	Grace Liu	Fusion Media	Basic Training Kit	Services
14	TXN-1014	45957	West	Daniel Cho	Fusion Media	Analytics Platform	Software
15	TXN-1015	45957	North	Aisha Khan	Keystone Edu	Cybersecurity Shield	Software
16	TXN-1016	45957	South	Priya Nair	Lumen Telecom	Enterprise CRM Suite	Software
17	TXN-1017	45958	West	Sofia Rossi	Apex Corp	Business Laptop L5	Hardware
18	TXN-1018	45959	East	Grace Liu	Meridian Auto	Wireless Router W2	Hardware
19	TXN-1019	45961	West	Sofia Rossi	Granite Bank	Server Rack X900	Hardware
20	TXN-1020	45962	East	Grace Liu	BlueWave Ltd	Cybersecurity Shield	Software
21	TXN-1021	45962	West	Sofia Rossi	Ionic Energy	Cloud Backup Pro	Software
22	TXN-1022	45962	North	Aisha Khan	Juniper Foods	Business Laptop L5	Hardware
23	TXN-1023	45964	South	Priya Nair	Granite Bank	Onboarding Package	Services

Step 4: Sorting and Filtering High-Value Transactions

· Created a table with Transaction ID, Product, Region, Customer, Revenue ($), and Profit ($).

· Sorted Revenue ($) in descending order so the highest-value transactions appeared first.

· Applied a revenue filter of $32,400 or more. The workbook contains 32 transactions meeting this threshold.

· The highest-value transaction is TXN-1156 for Enterprise CRM Suite, with revenue of $48,000.

Step 5: Filtering Underperforming Products

· Used Product_Summary to review product revenue, profit, margin, return rate, and status.

· Filtered Status to Underperforming and sorted Revenue ($) in ascending order to show the lowest-revenue products first.

· The filtered list contains Basic Training Kit, Legacy Fax Gateway, Smart Display 55in, and Wireless Router W2.

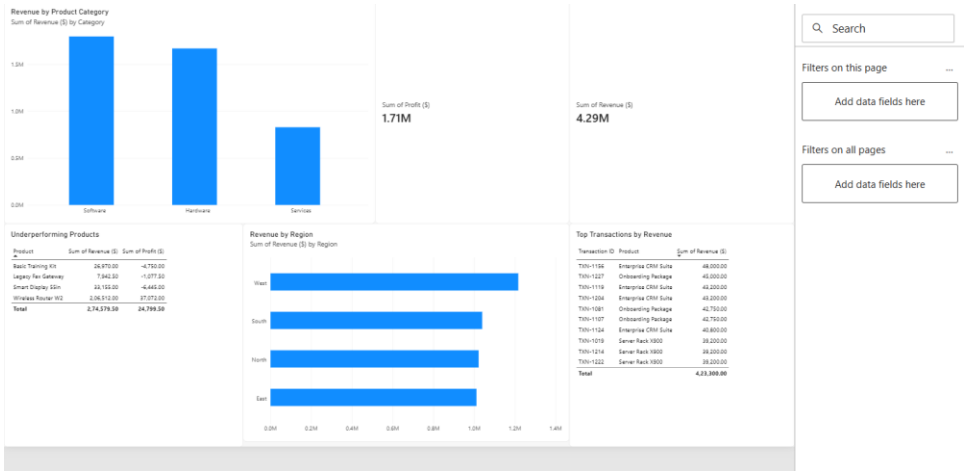
· The workbook flags products when revenue is below 60% of average product revenue or profit margin is below 15%.

Step 6: Dashboard Development

· Added summary cards for total revenue and total profit using Power BI built-in Sum aggregation.

· Created charts for revenue by product category and region, plus tables for underperforming products and top transactions by revenue.

· The submitted dashboard uses built-in field aggregations; custom DAX measures are not shown in the provided screenshot.



Output

The final Power BI report provides:

· Total Revenue of approximately $4.29M and Total Profit of approximately $1.71M.

· Revenue comparisons across Software, Hardware, and Services categories, and across the four regions.

· A descending list of high-value transactions, led by TXN-1156 at $48,000.

· A filtered list of four underperforming products for further review.

The report makes it easier to locate high-value transactions and identify products that may need attention.

Conclusion

This assignment used sorting and filtering in Power BI to examine high-value sales transactions and underperforming products. Cards, charts, and tables present revenue and profit totals, category and regional comparisons, and the products and transactions that require attention. These results support clearer sales review and business decisions.

Learning Outcomes

Through this assignment, I learned:

· How to import an Excel workbook and inspect data in Power Query.

· How to sort transactions by revenue in descending order.

· How to apply numeric and status filters to isolate relevant records.

· How to identify underperforming products using the provided summary status.

· How to present findings with cards, charts, and tables in a Power BI report.

Github - https://github.com/Syahrulnalli/Business-Analytics